# Data-Driven Modeling of Li-Ion Battery Based on the Manufacturer Specifications and Laboratory Measurements

Roberta Di Fonso, *Member, IEEE*, Carlo Cecati, *Life Fellow, IEEE*, Remus Teodorescu, *Fellow, IEEE*, Daniel-Ioan Stroe, *Member, IEEE*, and Pallavi Bharadwaj, *Member, IEEE*

***Abstract*—Accurate modeling of Lithium-ion battery is essential in the development and testing of state estimation and lifetime prediction algorithms. The desired features of the model include flexibility, fast development, accuracy and reliability. There are many different ways to model a battery, depending on the level of abstraction desired, the data available and the target application environment. This paper shows how to extract equivalent circuit model parameters from manufacturer datasheets and laboratory measurement to build robust battery simulation models. A step-by-step methodology for data preparation is presented for both datasheet and measurement-based methods. The benefits and the disadvantages of both approaches are also discussed. A simple equivalent circuit model is firstly derived from manufacturer specification and its robustness is enhanced by collecting more extensive experimental data in the laboratory. Furthermore, an advanced model to better capture the battery dynamics is developed. The aging effects are added to this battery model, to reflect the internal parameters variation according to the health condition of the battery. To measure the accuracy of the developed models, the relative error is computed. An initial relative error of 2.8% of the model build with manufacturer specifications is reduced to 1.0% using laboratory measurements and finally to less than 0.4% by incorporating aging effects.**

***Index Terms*—Lithium-ion battery, equivalent circuit model, datasheet specifications, data driven modeling, state of health.**

## I. Introduction

LI-ION batteries are the preferred energy source for the ever-expanding world of mobile applications, ranging from low-power in smartphones to high-power in electric vehicles (EVs) [1]. Batteries are also widely employed in renewable energy integration where they can act as energy storage and mitigate the grid dynamics in transient conditions [2], [3]. Proper battery modeling is fundamental to optimize energy flow for the battery, renewable energy resources, and the grid for high battery performance, energy dispatch, and reliability [4]. Plug-in electric vehicles will be pervasive in the upcoming years, it is therefore imperative to understand the load behavior of EV charging systems. A high number of vehicles charging from a distribution grid at the same time could have negative consequences on the distribution system. Therefore efficient EV load models need to be developed in order to investigate the potential effects on distribution grid security and functionality [5], [6]. Such a wide spectrum of systems requires quite different battery models, based on different chemistry and with different behavior at the electrical terminals. Batteries are complex non linear systems that must be run under the supervision of a BMS (Battery Management System) in order to assure safe charge/discharge operations. The development of a complete autonomous system requires fine-tuning of many algorithms, which include the state of charge (SOC) and state of health (SOH) estimation algorithms using voltage and currents at the battery terminals [7], [8]. Thus precise simulations of various components play a crucial role in the successful design of an efficient BMS. Proper modeling is crucial to understand how the BMS interacts with the battery pack and to build and represent the starting point of building more complete simulations of complex usage scenarios. For instance, the model could contain a battery pack connected to a photovoltaic panel to develop and test BMS algorithms based on the changing operating conditions [9], while in an EV, interior permanent magnet synchronous motor could represent the electronic load during standard drive cycles. A good battery model is a fundamental step for the development of innovative BMS performance optimization algorithms [10], [11]. The simulation of the battery allows to gain insight into the dynamic behavior of the battery pack and allows to explore estimation and balancing algorithms, as well as testing operational cases. Different field applications require different aspects of battery modeling to be highlighted based on aging conditions. Therefore, a robust battery model is required for field applications. Depending on the desired level of abstraction, a multitude of models have

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been developed by researchers worldwide. There are three big categories of battery models: (i) Physics and multi-physics models that aim at describing the complex internal chemical phenomena of batteries [12]; these can be considered lower-level models (ii) empirical models that use past experimental data to develop a strategy for predicting future behavior [13]; (iii) Equivalent circuit models (ECM) which aim at simulating the battery behavior using circuit elements [14], these models can be considered to be at higher level. Physics models are the most accurate since they can capture the internal electrochemical behavior of the cell. The most famous among these models is the Doyle-Fuller-Newman model, which describes the battery internal states using sets of partial differential equations. These equations require a high number of parameters that are usually determined using complex and time-consuming experiments that usually involve cycling of a cell in the laboratory [15]. The literature on the Doyle-Fuller-Newman model is vast and researchers continue to put effort both in optimizing the estimation of the model parameters [16] and in developing software solutions for making the model perform faster [17]. For system-level applications ECM models are a computationally-lighter alternative to physics-based models. The behavior of the cell is described by common circuit elements whose properties represent the internal chemical processes. ECM are usually static models, but recently researchers started to consider dynamic circuits, that use variable elements tuned either by look-up tables or by artificial intelligence techniques [18], [19], [20], [21], [22]. The optimal choice of the battery model comes down to three elements: (i) ready to use in a small amount of developing time, (ii) tailored to a specific battery of interest, and (iii) a good trade-off between accuracy and complexity. The ECM parameters are unique to each battery's chemistry, requiring to be determined using test data, targeting the application of interest. Collecting experimental data in the laboratory also allows to perform experiments at different aging times: information that is crucial for developing a model which adheres to reality. Having a model that can replicate how the battery parameters change according to usage and aging could be used to develop BMS algorithms for battery optimization. For instance, in EVs observing a change in battery parameters could signal a power fade, i.e., drive cycle loss [23]. This information could then be used to operate the battery optimally with the goal of extending the driving range or to develop energy-aware charging strategies. The possibility of observing the battery capacity fade can also allow seeing when it is time to go into the 2nd-lifetime application. In the Matlab Simulink environment, there are some ready-to-use library blocks that simulate a custom battery at different levels of accuracy by configuring some ECM parameters. The Datasheet-Battery-Block from Simulink implements a simplified model that requires only vectors of OCV and series resistance. These vectors can be extracted from the datasheets provided by the battery manufacturer [24], [25]. Building a Simulink model directly from the commercial datasheet is very useful and relatively fast. This data, however, is not ready to use but needs to be extracted in a precise way in order to be used successfully, for which a systematic methodology is presented in this paper. However, since standard procedures do not exist, the datasheets show data at discharge C-rates and temperatures chosen by the manufacturer. These datasets could thus be very different from the ones that would be used in real-life applications of the battery. In order to further enhance the robustness of the battery model for real field applications, additional experimental data is collected in the laboratory and used in further training of the datasheet model: these controlled experiments allow to incorporate additional conditions of interest as are faced by a battery in field applications.

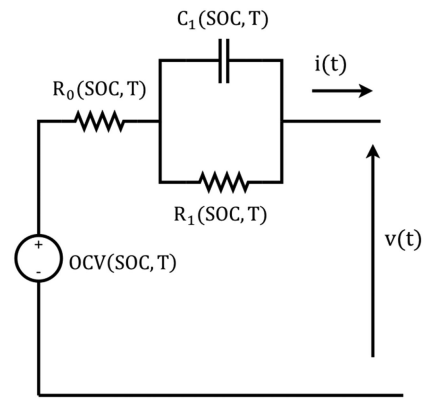


Fig. 1. Equivalent circuit model at a given battery age.

This paper presents a methodological approach of robust battery modelling whose complexity depends on the application at hand. The contributions of this paper are: (1) Development of a simple ECM of Li-ion battery using datasheet specification from fundamental; (2) Enhancement of datasheet model with planned laboratory measurements; (3) Development of a dynamic battery model with pulsed discharge data; (4) Extension of the model to consider battery aging effects; (5) Bench-marking the benefits and challenges of the different modelling methodologies. (6) Replication of results with publicly available datasets to enable researchers worldwide to develop a complete battery model regardless of having access to a battery laboratory; (7) This robust battery model finds wide application such as battery cell simulators for testing of Battery Management Systems in Hardware-in-the-Loop system [26], in EV performance optimization, etc.

## II. Equivalent Circuit Model

Li-ion batteries are based on complex chemistry whose understanding is not easy. However, from an application point of view, only the battery behavior at the accessible terminals is of interest. As an example, it is necessary to know the instantaneous output voltage when the external load requires current. This phenomenological behavior can be captured by the ECM containing common electrical elements. The complexity of the ECM model can be tailored to the required simulation precision, but the simple first-order circuit shown in Fig. 1 is already useful for many purposes. As it is possible to observe, the parameters of the ECM depend on temperature and on SOC. A complementary definition of the SOC is the depth of discharge (DOD) which represents the amount of charge that has already

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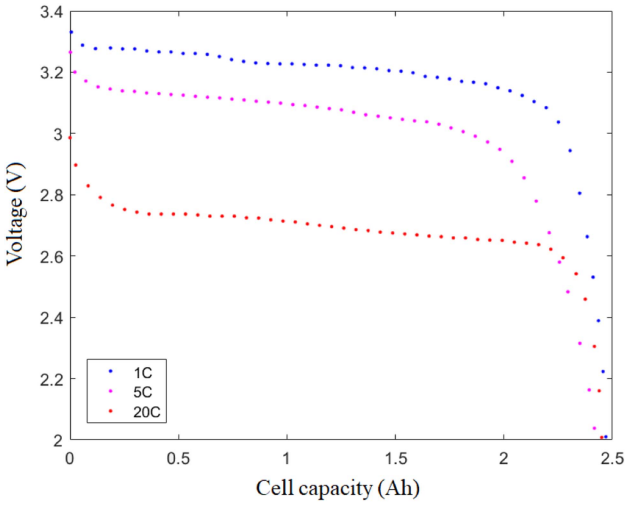


Fig. 2. Imported raw vectors for voltage vs. cell capacity at 23 °C.

been drawn from the battery. The relation between SOC and DOD is the following: DOD=1-SOC. The OCV generator is the main non-linear element as it depends heavily on SOC and on temperature. The RC elements depend as well on the same variables but, as a first approximation, can be considered fixed. The RC elements can be related to the physics inside the battery: the IR $R_0$ simulates the electrolyte resistance; $R_1$ simulates the charge-transfer resistance over the electrode-electrolyte interface; $C_1$ simulates the double-layer capacitance due to charges building up at the electrode surface. To develop an accurate model of a specific battery the corresponding OCV is needed along with the equivalent RC elements, all as a function of the working state of the battery, namely SOC and SOH which further defines the battery aging conditions [14].

## III. Datasheet Battery Block

One of the most readily available and basic tools of battery modeling is found in Matlab, called the Datasheet Battery Block [27] which is contained in the Simulink Library and simulates a Li-ion, lithium-polymer, or lead-acid battery cell. The model can be customized to the type of battery of interest, by using the manufacturer data. Internally the block is built using an ECM, as shown in Fig. 1. The inputs of the datasheet battery block are the battery initial charge capacity rated at nominal temperature, the battery current, and the battery temperature. The output voltage is computed using look-up tables for the open circuit voltage (OCV) and the internal resistance (IR). These look-up tables are user-generated by using the discharge characteristics provided by the manufacturer. By using a standard procedure, the user can parameterize the OCV and IR depending on SOC and temperature, and thus the battery performance is characterized at different operating points. The Datasheet Battery Block can either represent a single cell, or a battery by combining cells in parallel or in series. As previously mentioned, the internal structure of the block is an ECM and this can be either a basic model, which just considers OCV and IR, as shown in this Section, or a more complex model that also considers one RC-branch. This can be done by selecting the option filtered in the output battery voltage box and choosing the RC time constant.

### A. Look-Up Table Generation Procedure

In order to get the Datasheet Battery Block model running, it is needed to create the parameters to fill up the look-up tables for the OCV and IR that will be used by the block. The procedure is broken down into steps:

- Import the datasheet data;
- Use curve-fitting techniques to obtain OCV and IR data at different temperatures;
- Validate the model by comparing the simulated values to the datasheet data and verifying the behavior of IR;
- Specify the Datasheet Battery Block parameters: rated capacity at nominal temperature, OCV table data, OCV breakpoints, IR table data, battery temperature breakpoints, battery capacity breakpoints and initial battery charge.

## IV. Extracting Data From Datasheet

The Simulink Datasheet Battery Block is a ready-to-use subsystem that can simulate an ECM model, customized with look-up tables. In particular, this block must be configured with the cell OCV and the cell internal resistance, both as a function of SOC and temperature T. Unfortunately, these data are not directly available from the cell manufacturers but can be extracted from the official datasheets, mainly provided in graphic form. Therefore there is the need to extract the numerical data in some way. This data recovery process from graphs is not easy and prone to errors. Many tools have been developed over the years to aid this process, and some of them are freely available online [28]. The sample points of interest can be extracted semi-automatically drawing a mask to isolate the curve of interest.

Fig. 2 shows the raw vectors imported in Matlab (i.e., the values extracted from the voltage vs. capacity curves using this procedure). The data are from the datasheet of A123 batteries recorded at 23 °C. The raw vectors are sampled semi-automatically and some points must be hand adjusted in the x and y coordinates. So the sampling period is not uniform and, moreover, the curves are not sampled synchronously. The synchronicity is needed in order to relate the curves algebraically. The solution is provided by the Matlab Interp1 function, which can interpolate and resample the curves. Fig. 3(a),(b) show the results after up-sampling to 101 points and lightly filtering the curves. The manufacturer provides two families of curves:

- Curves at constant temperature and different discharge currents (Fig. 3(a));
- Curves at constant current and different temperatures (Fig. 3(b)).

The OCV curve at zero current is needed. In order to extract this information, a surface can be fit to the curves of Fig. 3(a), obtaining Fig. 4 The black curves are the original ones from Fig. 3(a), at discharge currents 1 C, 5 C and 20 C. Note that the

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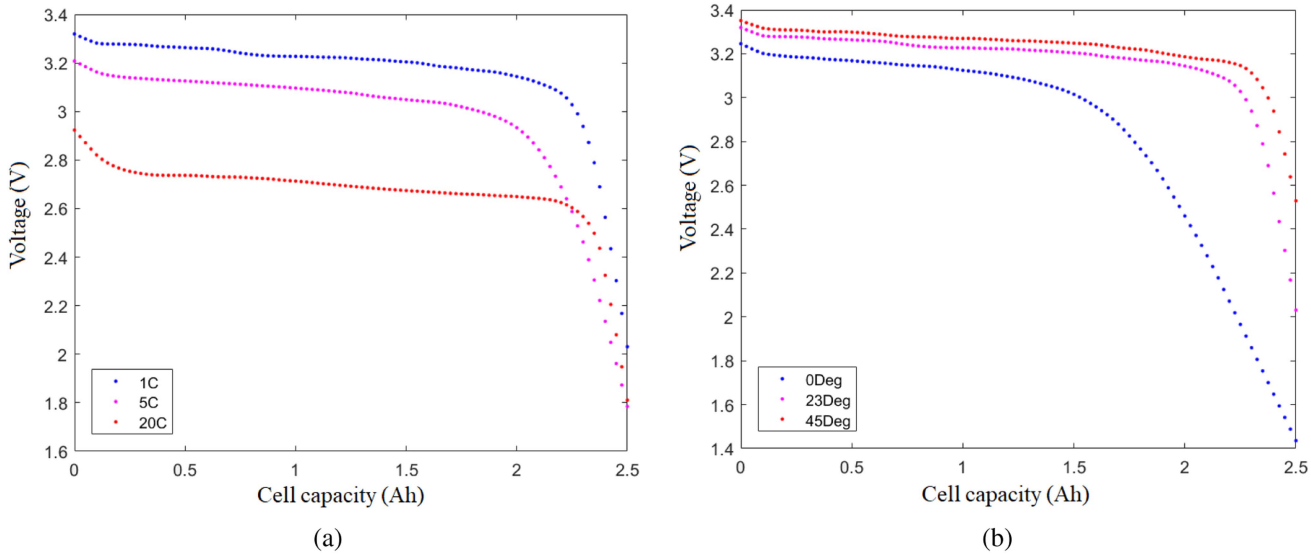


Fig. 3. Resampled voltages vs. capacity curves: (a) 23 °C and different C-rates, (b) 1C and different temperatures.

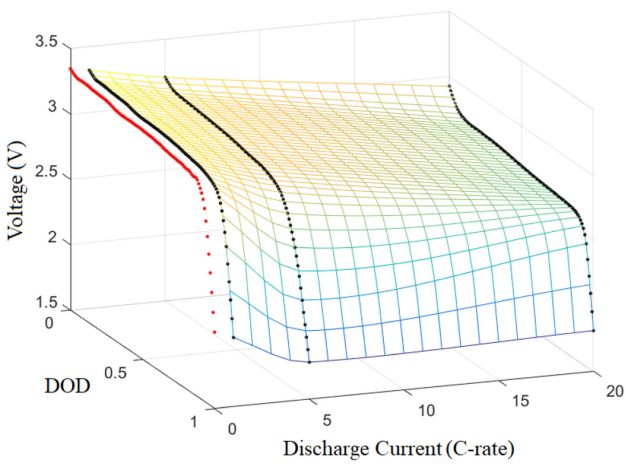


Fig. 4. Voltage surface vs. discharge current and DOD.

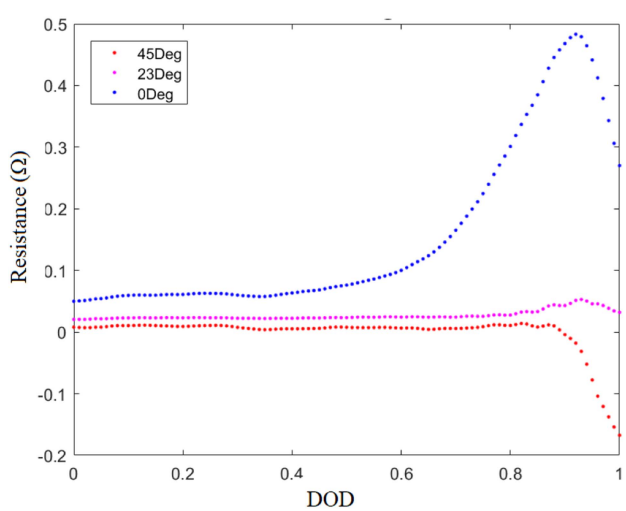


Fig. 5. $R_0$ computation from datasheet data.

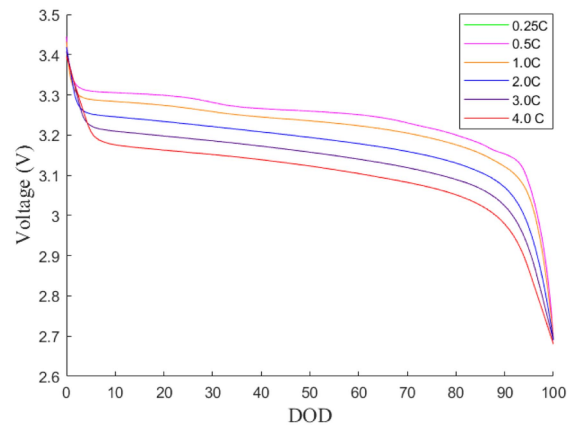


Fig. 6. Laboratory collected data showing discharge voltage vs. SOC at 25 °C at different C-rates.

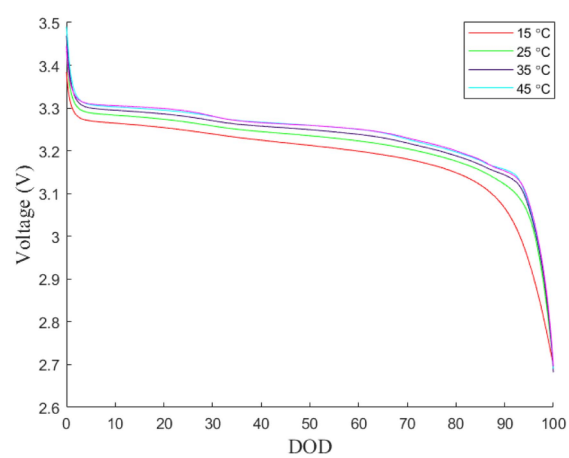


Fig. 7. Laboratory collected data showing discharge voltage at 1 C at different temperatures.

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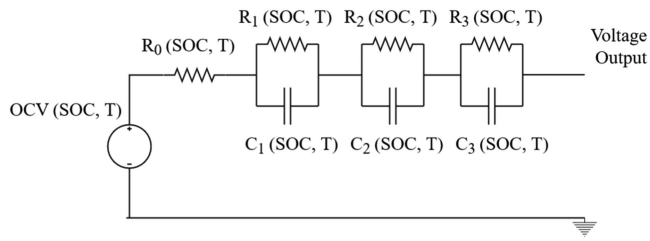


Fig. 8. Equivalent circuit battery block ECM with N= 3.

surface can now be cut at any desired value between 1 C and 20 C. However, to obtain the surface cut at zero current, i.e., the OCV curve, the surface has to be extrapolated. The solution is provided by the Matlab function Interp2, using the extrapolation method spline. The final answer is the red curve of Fig. 4, which can be directly used to configure the $E_m$ parameter of the Datasheet Battery Block. From the OCV curve, it is now possible to compute the internal resistance $R_0$ at different temperatures, according to (1). The OCV is the red curve shown in Fig. 4, while V(T) are the data shown in Fig. 3(b). The curves can be related algebraically, element-wise, to obtain $R_0$ as a function of SOC and temperature.

$$\mathrm{R_0(SOC, T) = (OCV - V(T))/1C} \tag{1}$$

The result is shown in Fig. 5. Unfortunately, OCV can be computed only at 23 °C and not as a function of T, due to lack of information from the manufacturer as shown in Fig. 3(a). This can generate some inconsistencies at low values of cell SOC that can be overcome by holding the value of $R_0$ constant in the range $0\% \leq \mathrm{SOC} \leq 10\%$. In Fig. 5 it is shown that, at very low SOC and temperature values, the internal resistance is high. This is in confirmation with the literature [29], [30] but at normal operation the internal resistance is low. In [31], internal resistance values for 2.5 Ah batteries at ambient temperature match those in Fig. 5.

## V. Extracting Data From Laboratory Data Collection

If laboratory equipment is available, it is possible to collect customized "datasheet like" data. The official data provided by the manufacturer are usually averaged on the characteristics of the batteries produced; furthermore, every company uses their own procedure, since standard procedures do not exist. Performing a data collection campaign in the lab has, therefore three main advantages:

1) the data are specific to the selected battery cell;
2) the ambient temperature and C-rates can be customized;
3) more extensive tests can be tailored to the final applications, including dynamic behavior and cycling aging.

The capacity of the A123 LFP-based Li-ion batteries was measured both during charging, using a standardized constant current – constant voltage (CC-CV) procedure, and discharging, using a CC procedure, at four different temperatures $T = 15\ ^\circ\mathrm{C},\ 25\ ^\circ C,\ 35\ ^\circ C,$. At each temperature, the battery capacity was measured at six C-rates $\mathrm{C-rates} = 0.25\mathrm{C},\ 0.5\mathrm{C},\ 1C,\ 2C,\ 3C,\ 4C$. Prior to each charging or discharging capacity measurement, the battery cell was kept for one hour at open circuit voltage condition, in order to ensure that it reached a thermo-dynamical stable condition. During all the measurements, the battery cell was placed into a climatic chamber to ensure stable and reliable temperature; furthermore, the mentioned temperature is the one measured on the surface of the cell using a type-k thermocouple.

The discharge voltage data at different C-rates and at a constant temperature of 25 °C are shown in Fig. 6 while the discharge voltage at constant 1 C and at different temperatures are shown in Fig. 7.

The same procedure has been performed in different aging states, with charge and discharge battery voltage profiles for temperature ranging from 15 °C to 45 °C and C-rates varying from 0.25 C to 4 C.

### A. Results

Using the battery manufacturer's data, it is possible to validate battery behavior for limited test conditions. However, using the advanced battery model with laboratory measurements, it is possible to replicate battery behavior under varied operating conditions. Fig. 9 shows the results of the datasheet battery block obtained by using the manufacturer's data. The A123 batteries have been tested by the manufacturer using two test cases: (1) Ambient temperature of 23 °C and $\mathrm{C-rates} = \{1C,\ 5C,\ 20C\}$; (2) 1 C rate and temperatures $T = \{0\ ^\circ\mathrm{C},\ 23\ ^\circ\mathrm{C},\ 45\ ^\circ\mathrm{C}\}$.

Fig. 9(a) shows the internal resistance as a function of SOC and temperature obtained using the set of temperatures available in the manufacturer's specification, while Fig. 9(b) shows the internal resistance as a surface. Fig. 10 shows the results of the datasheet battery block obtained using data collected in the lab. The A123 batteries have been tested at more operating conditions. In this case, the conditions are: (1) Ambient temperature of 25 °C and $\mathrm{C-rates} = \{0.25\mathrm{C},\ 0.5\mathrm{C},\ 1\mathrm{C},\ 2\mathrm{C},\ 3\mathrm{C},\ 4\mathrm{C}\}$; (2) 1 C rate and temperatures $\mathrm{T} = \{15\ ^\circ\mathrm{C},\ 25\ ^\circ\mathrm{C},\ 35\ ^\circ\mathrm{C},\ 45\ ^\circ\mathrm{C}\}$. Fig. 10(a) shows the obtained internal resistance as a function of SOC and temperature, while Fig. 10(b) shows internal resistance as a surface. In both Figs. 9 and 10 it is possible to observe that for lower temperatures, the internal resistance exhibits a higher value. Therefore, as the temperature decreases, the internal resistance of the battery shows an increasing trend, as shown in the surfaces of Figs. 9(b) and 10(b). A thorough battery behavior can be studied in the laboratory using which a more robust model can be made, as shown in Fig. 10.

To compare the two models, it has been computed the relative error RE between the voltage curves used to build the datasheet battery block and the voltage output of the block once it has been parametrized following the procedure shown in this Section. Equation (2) shows how to compute the relative error.

$$RE = \frac{(V_{comp} - V_{exp})}{V_{exp}} \cdot 100 \tag{2}$$

$V_{exp}$ represents either the data extracted from the manufacturer's datasheet or the laboratory measurements, while $V_{comp}$ is the output of the datasheet battery block. The RE represents the

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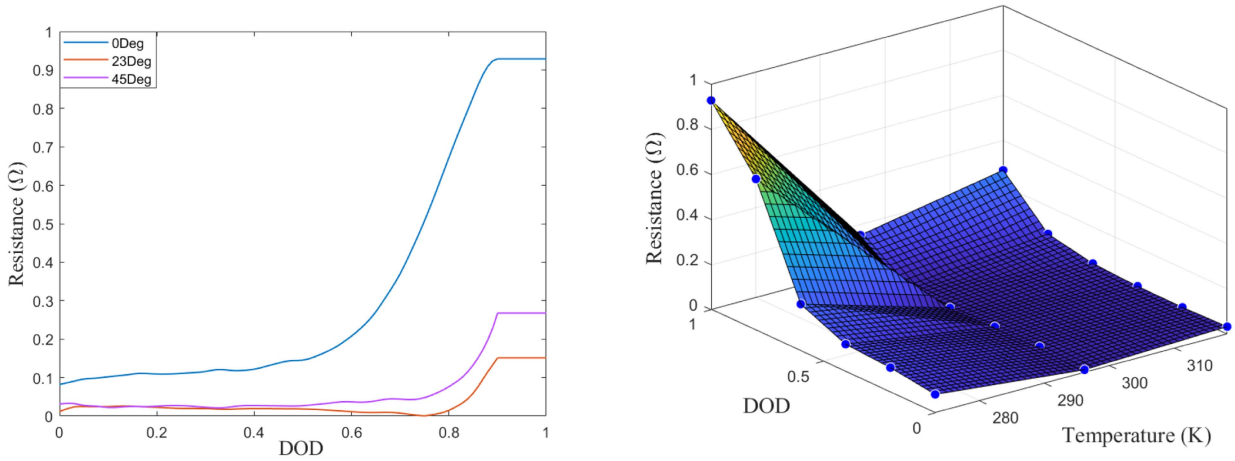
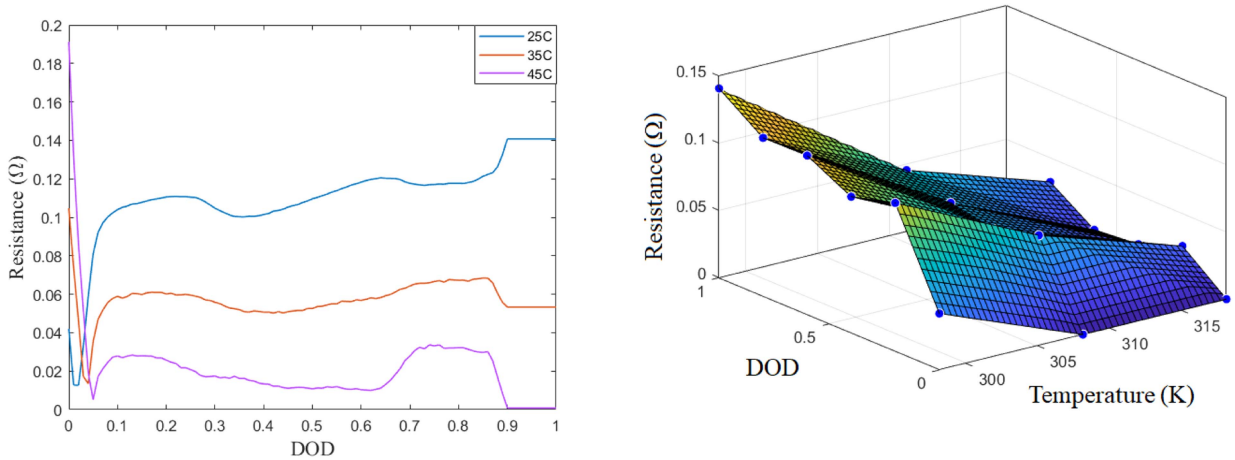



(a) (b)

Fig. 9. Datasheet model based on manufacturer's specifications: (a) internal resistance, (b) internal resistance surface.




(a) (b)

Fig. 10. Advanced datasheet battery model using laboratory measurement: (a) internal resistance, (b) internal resistance surface.

accuracy of the battery model output, compared to the experimental voltages.

The datasheet battery block built using data extracted from the manufacturer data has RE $\leq 2.8\%$ for $10\% \leq$ SOC $\leq 100\%$, while the battery block built using customized laboratory measurements has a RE $\leq 1.0\%$ for $10\% \leq$ SOC $\leq 100\%$.

The internal resistance shows an increasing trend as the temperature decreases. The effect of aging in battery modeling will be shown in the next Section.

## VI. Equivalent Circuit Battery Block

While a simple ECM can be very useful in many applications, a more complex ECM with a higher number of RC branches can help better capture the internal dynamics of the battery. The equivalent circuit battery block from Simulink implements an ECM model with a number of RC pairs that range from 1 to N [32]. In this work, N= 3, leading to the ECM model shown in Fig. 8. This ready-to-use block provides an easy-to-follow procedure that allows the automatic generation of ECM parameters by leveraging MathWorks tools, estimation techniques, and measured laboratory data. The procedure estimates parameters to fit a model to experimental pulse discharge signals, provides initial estimates for the ECM parameters, and computes the number of recommended RC pairs using numerical optimization techniques. The ideal data requirements are:

- Sample rate should be a minimum of 1 Hz, with an ideal rate at 10 Hz;
- Voltage accuracy within $\pm$ 5 mV, ideally within $\pm$ mV;
- Current accuracy within $\pm$ 100 mA, ideally within 10 mA;
- Change in SOC for each pulse should not be greater than 5%;
- Sufficient relaxation time after each pulse to ensure the battery approaches steady-state voltage.

The pulsed voltage discharge sequences are used for the pulse identification. To determine the number of necessary RC time

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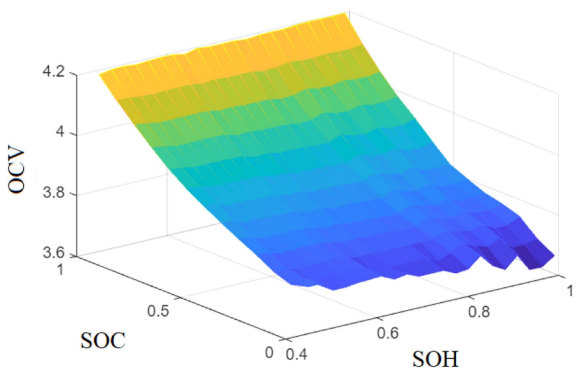
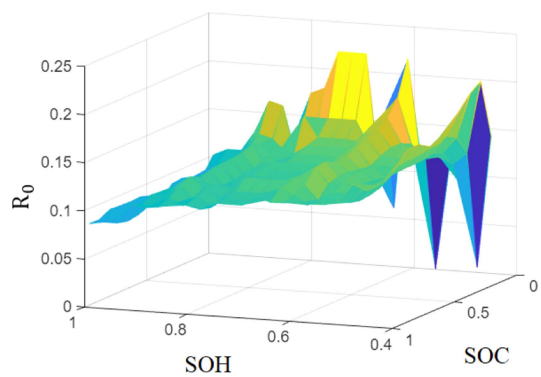
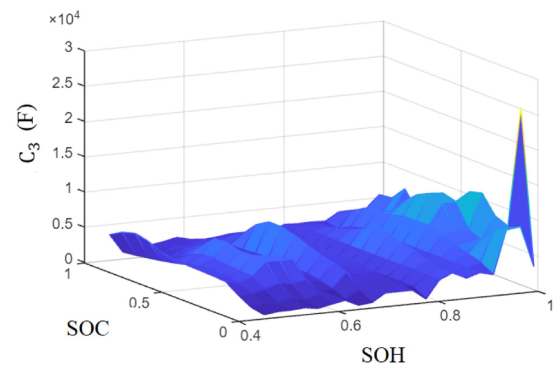



(a)

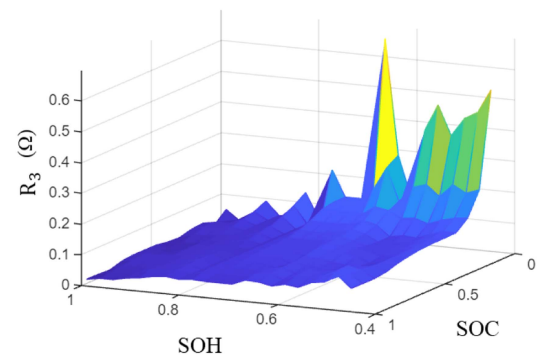



(b)

Fig. 11. ECM parameters values: (a) OCV (SOC,SOH), (b) $R_0$(SOC,SOH).




(a)




(b)

Fig. 12. ECM parameters values: (a) $C_3$(SOC,SOH), (b) $R_3$(SOC,SOH).

constants, the procedure performs curve-fitting on the pulse relaxation time interval at each different SOC. The parameters are finally put into a set of vectors that will be used in the ECM Simulink block.

### A. Model Enhancement for Aging

By repeating the procedure on different input data sequences with different SOH, the model can be enhanced by adding another dimension: the aging state. In this way, the aging of the battery becomes an additional parameter that can be fed as input to the battery model. This model is named as aging-aware equivalent circuit battery model. This model could then be used either to simulate a capacity fade while the battery is working by providing the SOH degradation curve as input or to simulate the battery at a specific age. The battery degradation Li-ion public dataset released from NASA is used [33]. The additional benefit of using publicly available datasets is that researchers worldwide can quickly develop a real-life battery model to work with, even if they have no access to the costly and specialized equipment needed to perform experiments. If experimental data of a battery of interest are available, the model can be custom-made to a specific battery. The NASA dataset has 21 pulsed discharge sequences of cell voltage and current, recorded at different times and synchronously sampled at 1 Hz. A current pulse of 1 A is applied for 10 minutes, followed by 20 minutes of rest.

### B. Results

The Equivalent Circuit Battery Block allows us to build a more complex ECM model that is able to model more accurately the battery internal electrochemical processes, including slow diffusion processes. By applying pulsed discharge voltage and current sequences for different SOH, the model can be enhanced to take into account ECM parameter variation with respect to aging conditions. In this way, the aging state becomes an additional input to the equivalent circuit battery block. Figs. 11 and 12 show how the model parameters change with respect to SOC and SOH. Only $R_3$, $C_3$ are shown since they have the largest time constants and, therefore, the biggest impact on the voltage output. Fig. 13 shows the results of the aging-aware equivalent circuit battery model for different SOH values. Fig. 13(a) shows the output voltage of the battery at the beginning of life, corresponding to SOH $= 1$ and initial full charge capacity of 2.1 Ah, while Fig. 13(b) shows the output voltage of the battery at the end of life, corresponding to SOH $= 0.45$ and initial full charged capacity of 0.9 Ah. The RE has been computed according to (2) for each of the 21 sequences in the NASA dataset and has been averaged. The aging-aware equivalent circuit battery model has a RE $\leq 0.4\%$ for $10\% \leq$ SOC $\leq 100\%$.

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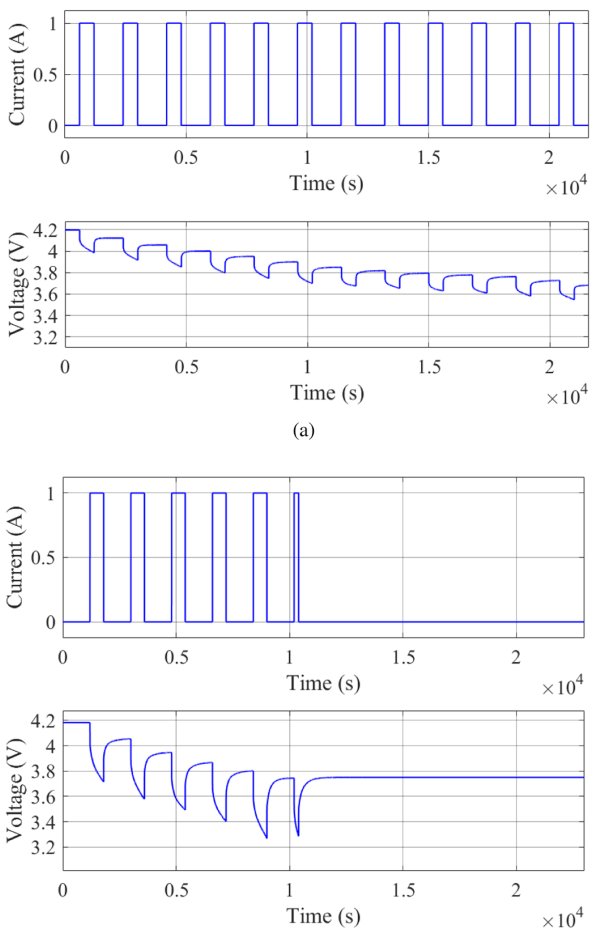


Fig. 13. Output voltage and current signals for different SOH: (a) SOH=1, (b) SOH=0.45.

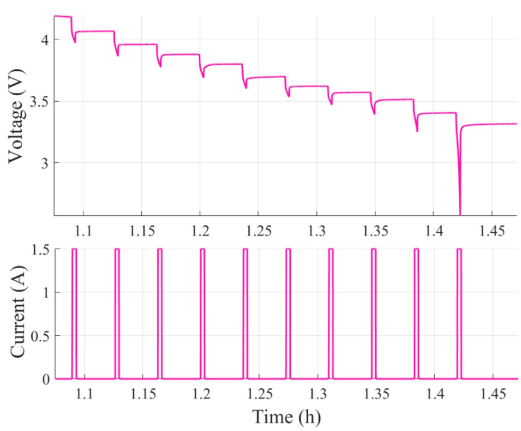


Fig. 14. Pulse power characterization test data.

### C. Results on Alternative Dataset

To show the validity the procedure show above, the same procedure is applied on another recent publicly available dataset, following the procedure shown in Section VI. The tested batteries are Panasonic NCR18650BD NCA cells, which are widely employed in electric vehicles battery packs available on the market [34]. The pulsed discharge data shown in Fig. 14 are from the pulse power characterization test on a NCA 3 Ah battery, with SOH= 90% [35].

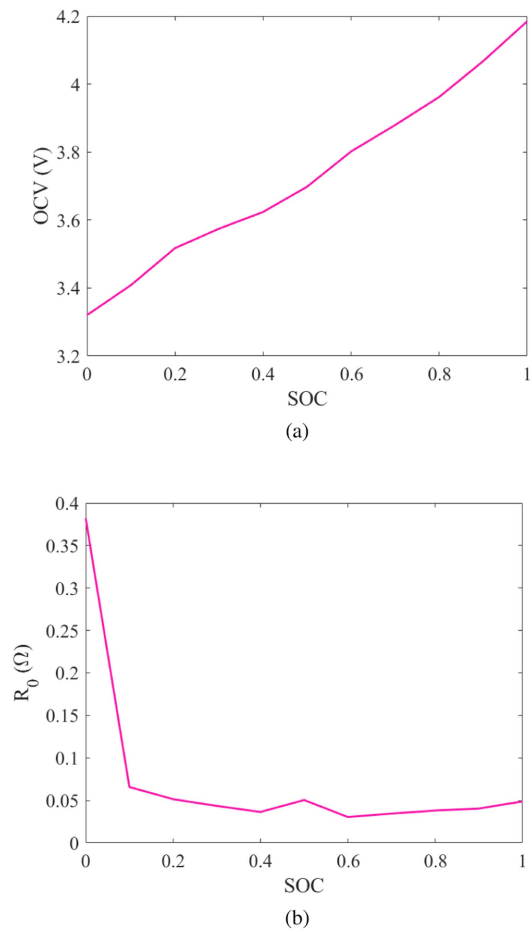


Fig. 15. ECM parameters values: (a) OCV(SOC) (b) $R_0$(SOC).

The RE, computed using (2) is $\mathrm{RE} \leq 0.4\%$ for $10\% \leq \mathrm{SOC} \leq 100\%$ The results for the $R_0$ and OCV parameters are shown in Fig. 15 and they are in accordance with the internal resistance and OCV from experimental tests provided in [34].

## VII. Conclusion

This paper presents a data-driven approach for modeling the behavior of a Li-ion battery. Firstly, it is shown how to use the manufacturer datasheets for configuring a simple equivalent circuit model using Matlab Simulink. To overcome the limited data from manufacturer datasheets, the battery model is tailored through data extracted from laboratory measurements for the field application of interest. This is achieved by using static discharge curves from two test cases: (1) ambient temperature of 25 °C and $\mathrm{C-rates} = \{0.25\mathrm{C},\ 0.5\mathrm{C},\ 1\mathrm{C},\ 2\mathrm{C},\ 3\mathrm{C},\ 4\mathrm{C}\}$; (2) 1 C rate and temperatures $\mathrm{T} = \{15\ ^\circ\mathrm{C},\ 25\ ^\circ\mathrm{C},\ 35\ ^\circ\mathrm{C},\ 45\ ^\circ\mathrm{C}\}$. Further, to capture the dynamic behavior of the battery it is shown how to build a more complex equivalent circuit model, is developed with a higher number of RC branches. The configuration data is extracted from pulsed discharge signals collected in the laboratory. The model is further enhanced to simulate

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the effects of the state of health of the battery. To showcase the replication of model with open source datasets, pulsed discharge is used from the repository of NASA and Oxford University. The proposed model shows high accuracy with an initial relative error of 2.8% for the model build with manufacturer specifications, which is reduced to 1.0% using laboratory measurements and finally to less than 0.4% by incorporating aging effects.

## REFERENCES


[1] R. Di Fonso, C. Cecati, R. Teodorescu, D.-I. Stroe, and P. Bharadwaj, "Data-driven modeling of li-ion battery based on the manufacturer specifications and laboratory measurements," in *Proc. 2022 IEEE Int. Conf. Power Electron., Drives Energy Syst.*, 2022, pp. 1–6.

[2] A. S. Subburaj, S. B. Bayne, M. G. Giesselmann, and M. A. Harral, "Analysis of equivalent circuit of the utility scale battery for wind integration," *IEEE Trans. Ind. Appl.*, vol. 52, no. 1, pp. 25–33, Jan./Feb. 2016.

[3] S. Pindado, J. M. Álvarez, E. Roibás-Millán, D. Alfonso-Corcuera, J. Cubas, and A. M. Gomez-San-Juan, "On the performance of solar cells/panels and Li-Ion batteries: Simplified models developed at the IDR/UPM institute," *IEEE Trans. Ind. Appl.*, vol. 57, no. 4, pp. 3969–3977, Jul./Aug. 2021.

[4] Z. N. Bako, M. A. Tankari, G. Lefebvre, and A. S. Maiga, "Experiment-based methodology of kinetic battery modeling for energy storage," *IEEE Trans. Ind. Appl.*, vol. 55, no. 1, pp. 593–599, Jan./Feb. 2019.

[5] A. M. Haidar and K. M. Muttaqi, "Behavioral characterization of electric vehicle charging loads in a distribution power grid through modeling of battery chargers," *IEEE Trans. Ind. Appl.*, vol. 52, no. 1, pp. 483–492, Jan./Feb. 2016.

[6] J. C. Gómez and M. M. Morcos, "Impact of EV battery chargers on the power quality of distribution systems," *IEEE Trans. Power Del.*, vol. 18, no. 3, pp. 975–981, Jul. 2003.

[7] R. Teodorescu, X. Sui, S. B. Vilsen, P. Bharadwaj, A. Kulkarni, and D.-I. Stroe, "Smart battery technology for lifetime improvement," *Batteries*, vol. 8, 2022, Art. no. 169.

[8] L. W. Juang, P. J. Kollmeyer, T. M. Jahns, and R. D. Lorenz, "Improved nonlinear model for electrode voltage–current relationship for more consistent online battery system identification," *IEEE Trans. Ind. Appl.*, vol. 49, no. 3, pp. 1480–1488, May/Jun. 2013.

[9] A. Berrueta, A. Soto, J. Marcos, Í. de la Parra, P. Sanchis, and A. Ursúa, "Identification of critical parameters for the design of energy management algorithms for Li-Ion batteries operating in PV power plants," *IEEE Trans. Ind. Appl.*, vol. 56, no. 5, pp. 4670–4678, Sep./Oct. 2020.

[10] A. Babin, N. Rizoug, T. Mesbahi, D. Boscher, Z. Hamdoun, and C. Larouci, "Total cost of ownership improvement of commercial electric vehicles using battery sizing and intelligent charge method," *IEEE Trans. Ind. Appl.*, vol. 54, no. 2, pp. 1691–1700, Mar./Apr. 2018.

[11] H. Bizhani, S. K. H. Sani, H. Rezazadeh, S. Muyeen, and S. Rahmani, "Modeling of an optimum fast charging multi-step constant current profile for lead-acid batteries," *IEEE Trans. Ind. Appl.*, vol. 59, no. 2, pp. 2050–2060, Mar./Apr. 2023.

[12] W. Gu and C. Wang, "Thermal-electrochemical modeling of battery systems," *J. Electrochem. Soc.*, vol. 147, no. 8, 2000, Art. no. 2910.

[13] E. Samadani et al., "Empirical modeling of lithium-ion batteries based on electrochemical impedance spectroscopy tests," *Electrochimica Acta*, vol. 160, pp. 169–177, 2015.

[14] R. Di Fonso, P. Bharadwaj, R. Teodorescu, and C. Cecati, "Internal resistance estimation of Li-ion batteries using wavelet analysis," in *Proc. IEEE 13th Int. Symp. Power Electron. Distrib. Gener. Syst.*, 2022, pp. 1–5.

[15] C. Chang-Hui, F. B. Planella, K. O'Regan, D. Gastol, W. D. Widanage, and E. Kendrick, "Development of experimental techniques for parameterization of multi-scale lithium-ion battery models," *J. Electrochem. Soc.*, vol. 167, no. 8, 2020, Art. no. 080534.

[16] Z. Khalik, M. Donkers, J. Sturm, and H. J. Bergveld, "Parameter estimation of the Doyle–Fuller–Newman model for Lithium-ion batteries by parameter normalization, grouping, and sensitivity analysis," *J. Power Sources*, vol. 499, 2021, Art. no. 229901.

[17] S. Moura, "Fast Doyle-Fuller-Newman (DFN) electrochemical-thermal battery model simulator, github." 2024. Accessed: Feb. 6, 2024. [Online]. Available: https://github.com/scott-moura/fastDFN.git

[18] R. Di Fonso, P. Bharadwaj, R. Teodorescu, and C. Cecati, "A battery digital twin based on neural network for testing SOC/SOH algorithms," in *Proc. IEEE 20th Int. Power Electron. Motion Control Conf.*, 2022, pp. 655–660.

[19] R. Di Fonso, R. Teodorescu, C. Cecati, and P. Bharadwaj, "A battery digital twin from laboratory data using wavelet analysis and neural networks," *IEEE Trans. Ind. Informat.*, vol. 20, no. 4, pp. 6889–6899, Apr. 2024.

[20] J. Hong et al., "Multi- forword-step state of charge prediction for real-world electric vehicles battery systems using a novel LSTM-GRU hybrid neural network," *eTransportation*, vol. 20, 2024, Art. no. 100322.

[21] R. Li, H. Wang, H. Dai, J. Hong, G. Tong, and X. Chen, "Accurate state of charge prediction for real-world battery systems using a novel dual-dropout-based neural network," *Energy*, vol. 250, 2022, Art. no. 123853.

[22] J. Hong, F. Liang, and H. Yang, "Research progress, trends and prospects of Big Data technology for new energy power and energy storage system," *Energy Rev.*, vol. 2, no. 3, 2023, Art. no. 100036.

[23] O. Theliander, A. Kersten, M. Kuder, W. Han, E. A. Grunditz, and T. Thiringer, "Battery modeling and parameter extraction for drive cycle loss evaluation of a modular battery system for vehicles based on a cascaded H-Bridge multilevel inverter," *IEEE Trans. Ind. Appl.*, vol. 56, no. 6, pp. 6968–6977, Nov./Dec. 2020.

[24] I. Korotkin, S. Sahu, S. E. O'Kane, G. Richardson, and J. M. Foster, "Dandeliion V1: An extremely fast solver for the Newman model of lithium-ion battery (dis) charge," *J. Electrochem. Soc.*, vol. 168, no. 6, 2021, Art. no. 060544.

[25] T. Huria, M. Ceraolo, J. Gazzarri, and R. Jackey, "High fidelity electrical model with thermal dependence for characterization and simulation of high power lithium battery cells," in *Proc. 2012 IEEE Int. Electric Veh. Conf.*, 2012, pp. 1–8.

[26] J. V. Barreras et al., "An advanced HIL simulation battery model for battery management system testing," *IEEE Trans. Ind. Appl.*, vol. 52, no. 6, pp. 5086–5099, Nov./Dec. 2016.

[27] "Lithium-ion, lithium-polymer, or lead-acid battery - simulink." 2024. Accessed: Feb. 6, 2024. [Online]. Available: https://www.mathworks.com/help/autoblks/ref/datasheetbattery.html

[28] WebPlotDigitizer - extract data from plots, images, and maps." 2024. Accessed: Feb. 6, 2024. [Online]. Available: https://automeris.io/WebPlotDigitizer

[29] X. Xu et al., "Flexible bidirectional pulse charging regulation achieving long-life lithium-ion batteries," *J. Energy Chem.*, vol. 96, pp. 59–71, 2024.

[30] S. Barcellona, S. Colnago, G. Dotelli, S. Latorrata, and L. Piegari, "Aging effect on the variation of Li-ion battery resistance as function of temperature and state of charge," *J. Energy Storage*, vol. 50, 2022, Art. no. 104658.

[31] J. Estaller, A. Kersten, M. Kuder, T. Thiringer, R. Eckerle, and T. Weyh, "Overview of battery impedance modeling including detailed state-of-the-art cylindrical 18650 lithium-ion battery cell comparisons," *Energies*, vol. 15, no. 10, 2022, Art. no. 3822.

[32] Generate parameter data for equivalent circuit battery block - MATLAB & simulink." 2024. Accessed: Feb. 6, 2024. [Online]. Available: https://www.mathworks.com/help/autoblks/ug/generate-parameter-data-for-estimations-circuit-battery-block.html

[33] B. Saha and K. Goebel, "Battery data set," NASA Ames Prognostics Data Repository, NASA Ames Res. Center, Moffett Field, CA, USA, 2007. [Online]. Available: http://ti.arc.nasa.gov/project/prognostic-data-repository

[34] T. Raj, "The impact of path dependent degradation on the lifetime of lithium-ion batteries," Ph.D. dissertation, University of Oxford, Oxford, U.K., 2021.

[35] T. Raj, "Dependent battery degradation dataset part 1," University of Oxford, 2020.


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